Differential Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.

Separate: $y\,dy = x\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y = 3, x = 0$: $\frac{9}{2} = C$. So $y^2 = x^2 + 9$, giving $y = \sqrt{x^2 + 9}$ (taking the positive root since $y = 3 > 0$).

2 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.

Separate: $\frac{dy}{y} = 2x\,dx$. Integrate: $\ln|y| = x^2 + C$. So $y = Ae^{x^2}$ for some constant A. Using $y = 5, x = 0$: $A = 5$. Solution: $y = 5e^{x^2}$.

3 **[Foundational]** • **Target: Modeling with differential equations**

A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k, to three decimal places.

Solving the DE: $P(t) = P_0e^{kt}$. Doubling: $2P_0 = P_0e^{5k}$, so $e^{5k} = 2$, giving $5k = \ln 2$, $k = \frac{\ln 2}{5} \approx 0.139$.

4 **[Foundational]** • **Target: Modeling with differential equations**

Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0.

Separate: $\frac{dT}{T - T_a} = -k\,dt$. Integrate: $\ln|T - T_a| = -kt + C$. So $T - T_a = Ae^{-kt}$. At $t = 0$, $T = T_0$, so $A = T_0 - T_a$. Solution: $T(t) = T_a + (T_0 - T_a)e^{-kt}$.

5 **[Foundational]** • **Target: First-order linear differential equations**

Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.

Integrating factor: $e^{\int 2\,dx} = e^{2x}$. Multiply through: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$, i.e. $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$. Integrate: $ye^{2x} = 3e^{2x} + C$, so $y = 3 + Ce^{-2x}$. Using $y = 1, x = 0$: $1 = 3 + C$, so $C = -2$. Solution: $y = 3 - 2e^{-2x}$.

6 **[Foundational]** • **Target: Euler's method for numerical solutions**

Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.

Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.5(0 + 1) = 1.5$. Step 2 ($x_1 = 0.5, y_1 = 1.5$): $y_2 = 1.5 + 0.5(0.5 + 1.5) = 1.5 + 1 = 2.5$. Estimate: $y(1) \approx 2.5$.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Second-order context: simple harmonic motion

The differential equation $\frac{d^2x}{dt^2} = -\omega^2 x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.

$\frac{dx}{dt} = -A\omega\sin(\omega t)$. $\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t) = -\omega^2(A\cos(\omega t)) = -\omega^2 x$, confirming $x(t) = A\cos(\omega t)$ satisfies the differential equation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Family of solution curves and particular solutions

Find the general solution of $\frac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.

Integrating directly: $y = x^3 + C$. The constant C represents a vertical shift — the general solution is a whole family of curves, all with the same shape (since they share the same derivative $3x^2$ at every x), just shifted up or down by different amounts.

9 [Intermediate] • Target: Family of solution curves and particular solutions

A tank drains such that $\frac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.

Separate: $\frac{dV}{\sqrt{V}} = -k\,dt$, i.e. $V^{-1/2}\,dV = -k\,dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $2\sqrt{V_0} = C$. So $2\sqrt{V} = -kt + 2\sqrt{V_0}$, giving $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, so $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

10 [Intermediate] • Target: Verifying a solution to a second-order differential equation

Verify that $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

$\frac{dy}{dx} = 2e^{2x}$. $\frac{d^2y}{dx^2} = 4e^{2x}$. Substituting: $4e^{2x} - 5(2e^{2x}) + 6e^{2x} = 4e^{2x} - 10e^{2x} + 6e^{2x} = 0e^{2x} = 0$, confirming $y = e^{2x}$ is a solution.

11 [Intermediate] • Target: Slope fields and qualitative solutions

A differential equation is given by $\frac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.

At $(0, 0)$: gradient $= 0 - 0 = 0$. At $(1, 0)$: gradient $= 1 - 0 = 1$. At $(1, 1)$: gradient $= 1 - 1 = 0$. A slope field is a diagram that plots a short line segment of the correct gradient at many points in the plane, without solving the equation explicitly — sketching a solution curve then amounts to drawing a curve that follows the direction of the segments at every point it passes through.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Solving a linear differential equation with an initial condition

Solve $\frac{dy}{dx} - \frac{y}{x} = x$ for $x > 0$ using an integrating factor, given $y = 2$ when $x = 1$.

Rewrite in standard form: $\frac{dy}{dx} - \frac{1}{x}y = x$. Integrating factor: $e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$. Multiply through: $\frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 1$, i.e. $\frac{d}{dx}\left(\frac{y}{x}\right) = 1$. Integrate: $\frac{y}{x} = x + C$, so $y = x^2 + Cx$. Using $y = 2, x = 1$: $2 = 1 + C$, so $C = 1$. Solution: $y = x^2 + x$.

13 [Intermediate] • Target: A logistic growth model

A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10-P)} = \frac{1}{10P} + \frac{1}{10(10-P)}$, solve for P in terms of t, given $P(0) = 1$ and leaving k as an unknown constant.

Separate: $\frac{dP}{P(10-P)} = k\,dt$. Integrate the left side using the given decomposition: $\frac{1}{10}\ln|P| - \frac{1}{10}\ln|10 - P| = kt + C$, i.e. $\frac{1}{10}\ln\left|\frac{P}{10-P}\right| = kt + C$. Using $P(0) = 1$: $\frac{1}{10}\ln\left(\frac{1}{9}\right) = C$. So $\ln\left|\frac{P}{10-P}\right| = 10kt + \ln\left(\frac{1}{9}\right)$, giving $\frac{P}{10-P} = \frac{1}{9}e^{10kt}$. Solving for P: $9P = (10 - P)e^{10kt}$, so $P(9 + e^{10kt}) = 10e^{10kt}$, giving $P = \frac{10e^{10kt}}{9 + e^{10kt}}$.

14 [Intermediate] • Target: Checking the order and degree of a differential equation

State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.

The order of a differential equation is the order of the highest derivative present. (a) Only a first derivative appears, so this is a first-order differential equation. (b) The highest derivative is $\frac{d^2y}{dx^2}$, so this is second-order. (c) The highest derivative is $\frac{d^3y}{dx^3}$, so this is third-order.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.

Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.25(2 \cdot 0) = 1 + 0 = 1$. Step 2 ($x_1 = 0.25, y_1 = 1$): $y_2 = 1 + 0.25(2 \cdot 0.25) = 1 + 0.125 = 1.125$. Estimate: $y(0.5) \approx 1.125$. Exact solution: $y(0.5) = (0.5)^2 + 1 = 1.25$. The Euler estimate underestimates the true value, since the method uses the derivative only at the start of each interval rather than accounting for how the derivative changes within it.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 4.02 The Poisson Distribution

Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.

Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$ for some constant A. Using $y = 6, x = 2$: $6 = 2A$, so $A = 3$. Solution: $y = 3x$. This represents the family of straight lines through the origin; the particular solution satisfying the initial condition is the specific line with slope 3.

17 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.01 Further Integration Techniques

A cup of coffee at $90\,^\circ$C is placed in a room at ambient temperature $20\,^\circ$C. Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to $60\,^\circ$C, find the value of k to three decimal places.

Using $T_a = 20$, $T_0 = 90$, and $T(10) = 60$: $60 = 20 + (90 - 20)e^{-10k}$, so $40 = 70e^{-10k}$, giving $e^{-10k} = \frac{4}{7}$. Taking natural logs: $-10k = \ln\left(\frac{4}{7}\right) \approx -0.5596$, so $k \approx 0.056$.

18 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.

(a) Salt flows in at a rate of 3 grams/litre $\times 2$ litres/minute $= 6$ grams/minute. Salt flows out at a rate proportional to the current concentration $\frac{S}{200}$ grams/litre, times the outflow rate of 2 litres/minute, giving $\frac{2S}{200} = \frac{S}{100}$ grams/minute out. Net rate: $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) This is a linear first-order DE. Integrating factor: $e^{\int \frac{1}{100} dt} = e^{t/100}$. $\frac{d}{dt}(Se^{t/100}) = 6e^{t/100}$. Integrating: $Se^{t/100} = 600e^{t/100} + C$, so.

19 [Challenging] • Target: First-order linear differential equations • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A cubic equation with real coefficients $x^3 + 7x^2 + 15x + 25 = 0$ has $-1 + 2i$ as one root. Find the real root.

Final Answer: -5. Complex roots occur in conjugate pairs, so $-1 - 2i$ is also a root. Product of all three roots $= -d = -25$, and $(-1 + 2i)(-1 - 2i) = 5$, so the real root $= \frac{-25}{5} = -5$.

20 [Challenging] • Target: Euler's method for numerical solutions • Cross-Lesson: 6.01 Matrices (AI HL)

A cubic equation with real coefficients $x^3 - 10x^2 + 37x - 52 = 0$ has $3 + 2i$ as one root. Find the real root.

Final Answer: 4. Complex roots occur in conjugate pairs, so $3 - 2i$ is also a root. Product of all three roots $= -d = 52$, and $(3 + 2i)(3 - 2i) = 13$, so the real root $= \frac{52}{13} = 4$.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$ for $y > 0$.

a. $y = \sqrt{2x^2 + 16}$ ✓

b. $y = 2x^2 + 4$

c. $y = \sqrt{x^2 + 16}$

d. $y = x^2 + 4$

Separate variables: $ydy = 2xdx$. Integrate: $\frac{1}{2}y^2 = x^2 + C \Rightarrow y^2 = 2x^2 + 2C$. Since $y(0) = 4$, $16 = 2C$. Thus $y^2 = 2x^2 + 16 \Rightarrow y = \sqrt{2x^2 + 16}$.

22 [Foundational] • Target: Separation of Variables

Solve $\frac{dy}{dx} = -3y$ with $y(0) = 50$.

a. $y = 50e^{-3x}$ ✓

b. $y = 50e^{3x}$

c. $y = -3x + 50$

d. $y = 50 - 3e^x$

$\frac{1}{y}dy = -3dx \Rightarrow \ln|y| = -3x + C \Rightarrow y = Ae^{-3x}$. With $y(0) = 50$, $y = 50e^{-3x}$.

23 [Foundational] • Target: Separation of Variables

Solve $\frac{dy}{dx} = (1 + y^2)e^x$ with $y(0) = 0$.

a. $y = \tan(e^x - 1)$ ✓

b. $y = \arctan(e^x - 1)$

c. $y = \tan(e^x)$

d. $y = e^x - 1$

$\frac{1}{1+y^2}dy = e^x dx \Rightarrow \arctan y = e^x + C$. With $y(0) = 0$, $\arctan 0 = e^0 + C \Rightarrow 0 = 1 + C \Rightarrow C = -1$. Thus $\arctan y = e^x - 1 \Rightarrow y = \tan(e^x - 1)$.

24 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

For the linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, what is the integrating factor $I(x)$?

a. $I(x) = e^{\int P(x)dx}$ ✓

b. $I(x) = \int P(x)dx$

c. $I(x) = e^{-\int P(x)dx}$

d. $I(x) = e^{\int Q(x)dx}$

Multiplying by $I(x) = e^{\int P(x)dx}$ collapses the left-hand side into the product rule derivative $\frac{d}{dx}[I(x)y]$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

Find the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$ for $x > 0$.

a. x^3 ✓

b. $3\ln x$

c. e^{3x}

d. $\frac{1}{x^3}$

$I(x) = e^{\int (3/x)dx} = e^{3\ln x} = e^{\ln(x^3)} = x^3$.

26 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

Solve $\frac{dy}{dx} + 2y = 4$ with $y(0) = 5$.

a. $y = 2 + 3e^{-2x}$ ✓

b. $y = 2 + 5e^{-2x}$

c. $y = 4 + e^{-2x}$

d. $y = 2 - 3e^{-2x}$

$I(x) = e^{2x}$. Multiplying: $\frac{d}{dx}[e^{2x}y] = 4e^{2x} \Rightarrow e^{2x}y = 2e^{2x} + C \Rightarrow y = 2 + Ce^{-2x}$. Using $y(0) = 5 \Rightarrow 5 = 2 + C \Rightarrow C = 3$. Thus $y = 2 + 3e^{-2x}$.

27 [Foundational] • Target: Homogeneous Differential Equations (y = vx)

Which substitution transforms a homogeneous differential equation $\frac{dy}{dx} = f(y/x)$ into a separable equation?

a. $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ ✓

b. $y = v + x$

c. $v = xy$

d. $y = v^2x$

Let $v = y/x \Rightarrow y = vx$. By the product rule, $\frac{dy}{dx} = v + x\frac{dv}{dx}$, reducing the ODE to $v + x\frac{dv}{dx} = f(v)$.

28 [Foundational] • Target: Homogeneous Differential Equations (y = vx)

Classify the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$.

a. **Homogeneous of degree 0** ✓

b. First-order linear

c. Exact non-linear

d. Second-order Bernoulli

Dividing numerator and denominator by x^2 gives $\frac{1 + (y/x)^2}{2(y/x)}$, which is a function purely of y/x, classifying it as homogeneous.

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Homogeneous Differential Equations (y = vx)**

Using $y = vx$, what does $\frac{dy}{dx} = \frac{y}{x} + 1$ become?

a. $x\frac{dv}{dx} = 1$ ✓

b. $v + \frac{dv}{dx} = 1$

c. $x\frac{dv}{dx} = v + 1$

d. $\frac{dv}{dx} = x$

Substitute $\frac{dy}{dx} = v + x\frac{dv}{dx}$ and $y/x = v$: $v + x\frac{dv}{dx} = v + 1 \Rightarrow x\frac{dv}{dx} = 1$.

30 **[Intermediate]** • **Target: Numerical Euler's Method**

What is Euler's recurrence relation for approximating solutions to $\frac{dy}{dx} = f(x, y)$ with step size h?

a. $y_{n+1} = y_n + hf(x_n, y_n)$ ✓

b. $y_{n+1} = y_n + \frac{h}{2}f(x_n, y_n)$

c. $y_{n+1} = f(x_n + h, y_n)$

d. $y_{n+1} = y_n + h^2 f(x_n, y_n)$

Euler's method uses the tangent line approximation: $y_{n+1} = y_n + h\frac{dy}{dx}\Big|_{(x_n, y_n)} = y_n + hf(x_n, y_n)$.

31 **[Intermediate]** • **Target: Numerical Euler's Method**

Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with step size $h = 0.1$ to find y_1 at $x = 0.1$.

a. 1.10 ✓

b. 1.05

c. 1.21

d. 1.15

$y_1 = y_0 + h(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.10$.

32 **[Intermediate]** • **Target: Numerical Euler's Method**

Using the result above ($x_1 = 0.1, y_1 = 1.10$), find y_2 at $x = 0.2$ with $h = 0.1$.

a. 1.22 ✓

b. 1.20

c. 1.25

d. 1.31

$y_2 = y_1 + h(x_1 + y_1) = 1.10 + 0.1(0.1 + 1.10) = 1.10 + 0.1(1.20) = 1.10 + 0.12 = 1.22$.

© ClipSAT — clipsat.org | All Rights Reserved

33 **[Intermediate]** • **Target: Separation of Variables**

A cubic equation with real coefficients $x^3 + 6x^2 + 10x + 8 = 0$ has $-1 + i$ as one root. Find the real root.

a. -7

b. -1

c. -5

d. **-4** ✓

Complex roots occur in conjugate pairs, so $-1 - i$ is also a root. Product of all three roots $= -d = -8$, and $(-1 + i)(-1 - i) = 2$, so the real root $= \frac{-8}{2} = -4$.

34 **[Intermediate]** • **Target: Integrating Factor Method for First-Order Linear ODEs**

A cubic equation with real coefficients $x^3 - 8x^2 + 22x - 20 = 0$ has $3 + i$ as one root. Find the real root.

a. -2

b. 3

c. **2** ✓

d. 1

Complex roots occur in conjugate pairs, so $3 - i$ is also a root. Product of all three roots $= -d = 20$, and $(3 + i)(3 - i) = 10$, so the real root $= \frac{20}{10} = 2$.

35 **[Intermediate]** • **Target: Homogeneous Differential Equations (y = vx)**

A cubic equation with real coefficients $x^3 - 2x + 4 = 0$ has $1 + i$ as one root. Find the real root.

a. 2

b. 0

c. -6

d. **-2** ✓

Complex roots occur in conjugate pairs, so $1 - i$ is also a root. Product of all three roots $= -d = -4$, and $(1 + i)(1 - i) = 2$, so the real root $= \frac{-4}{2} = -2$.

36 **[Intermediate]** • **Target: Numerical Euler's Method**

A graph has edges $1 - 2, 2 - 3, 3 - 4, 4 - 1$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists ✓

b. Euler trail exists, but not an Euler circuit

c. Cannot be determined from the given information

d. Neither an Euler trail nor circuit exists

A connected graph has an Euler circuit iff every vertex has even degree; an Euler trail (not circuit) iff exactly 2 vertices have odd degree.

© ClipSAT — clipsat.org | All Rights Reserved

37 **[Intermediate]** • **Target: Solving by separation of variables**

A graph has edges $1-2, 2-3, 3-4, 4-1, 1-3$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Neither an Euler trail nor circuit exists

b. Euler circuit exists

c. Euler trail exists, but not an Euler circuit ✓

d. Cannot be determined from the given information

A connected graph has an Euler circuit iff every vertex has even degree; an Euler trail (not circuit) iff exactly 2 vertices have odd degree.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Modeling with differential equations** • **Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

A graph has edges $1-2, 2-3, 3-4, 1-4, 1-3, 2-4$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists

b. Neither an Euler trail nor circuit exists ✓

c. Cannot be determined from the given information

d. Euler trail exists, but not an Euler circuit

A connected graph has an Euler circuit iff every vertex has even degree; an Euler trail (not circuit) iff exactly 2 vertices have odd degree.

39 **[Challenging]** • **Target: First-order linear differential equations** • **Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

Factor $P(x) = x^3 + 7x^2 + 14x + 8$ completely. Find the largest root.

a. **-1** ✓

b. -2

c. -3

d. 0

Working through this additional-topics step: $P(x) = (x+2)(x+4)(x+1) \Rightarrow \text{roots} = -2, -4, -1;\ \text{largest} = -1$

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Euler's method for numerical solutions • Cross-Lesson: 6.01 Matrices (AI HL)

Factor $P(x) = x^3 + 4x^2 - 9x - 36$ completely. Find the largest root.

a. 0

b. **3** ✓

c. 1

d. 2

Working through this additional-topics step: $P(x) = (x-3)(x+4)(x+3) \Rightarrow \text{roots} = 3,\ -4,\ -3;\ \text{largest} = 3$

41 [Challenging] • Target: Second-order context: simple harmonic motion • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Factor $P(x) = x^3 + 2x^2 - x - 2$ completely. Find the largest root.

a. -2

b. 4

c. **1** ✓

d. 2

Working through this additional-topics step: $P(x) = (x-1)(x+2)(x+1) \Rightarrow \text{roots} = 1,\ -2,\ -1;\ \text{largest} = 1$

42 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Factor $P(x) = x^3 + 2x^2 - 11x - 12$ completely. Find the largest root.

a. **3** ✓

b. 5

c. 6

d. 4

Working through this additional-topics step: $P(x) = (x+4)(x+1)(x-3) \Rightarrow \text{roots} = -4,\ -1, 3;\ \text{largest} = 3$

43 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 1.01 Proof by Mathematical Induction

Factor $P(x) = x^3 - x^2 - 4x + 4$ completely. Find the largest root.

a. **2** ✓

b. 1

c. 4

d. 3

Working through this additional-topics step: $P(x) = (x-1)(x-2)(x+2) \Rightarrow \text{roots} = 1, 2,\ -2;\ \text{largest} = 2$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Factor $P(x) = x^3 - 2x^2 - 9x + 18$ completely. Find the largest root.

a. 0

b. 2

c. **3** ✓

d. 4

Working through this additional-topics step: $P(x) = (x + 3)(x - 3)(x - 2) \Rightarrow \text{roots} = -3, 3, 2;\ \text{largest} = 3$

45 [Challenging] • Target: Solving a linear differential equation with an initial condition • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Factor $P(x) = x^3 - 4x^2 - 4x + 16$ completely. Find the largest root.

a. 3

b. **4** ✓

c. 5

d. 6

Working through this additional-topics step: $P(x) = (x - 2)(x - 4)(x + 2) \Rightarrow \text{roots} = 2, 4, -2;\ \text{largest} = 4$

© ClipSAT — clipsat.org | All Rights Reserved